MTH208

Timed Online Assignment – July Semester 2020

Advanced Linear Algebra

Tuesday, 17 November 2020 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SEVEN (7)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, SUSS PI No., and Your Name. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Calculate the Jordan canonical form of the matrix.

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{pmatrix}$$

(15 marks)

Question 2

Let

$$w_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 2 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}, w_3 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

be linearly independent vectors in $\mathbb{R}^4$. Define $\mathcal{B} = \{w_1, w_2, w_3\}$ and let W be the subspace of $\mathbb{R}^4$ spanned by the ordered basis $\mathcal{B}$.

(a) Use the Gram-Schmidt orthonormalisation process to derive an orthonormal basis $\mathcal{C}$ of W.

(7 marks)

(b) A linear operator $T: W \to W$ is defined by

$$T(w_1) = w_1 + 3w_2$$

$$T(w_2) = w_1 - w_3$$

$$T(w_3) = 2w_2 - w_3.$$

Let T^* denote the adjoint of T with respect to the standard inner product. Compute $[T^*]_{\mathcal{C}}$, the matrix representation of T^* relative to the orthonormal basis $\mathcal{C}$ derived in (b).

(8 marks)

Question 3

Let

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 2 & -1 \end{pmatrix},$$

and let $B = A^*A$.

(a) Prove that $B = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ and that the eigenvalues of B are 3 and 9.

(3 marks)

(b) Demonstrate the property of normal operators by deriving a unitary matrix Q and a diagonal matrix D such that $Q^*BQ = D$.

(6 marks)

(c) Provide unitary matrices U, V and a matrix Σ with non-negative entries satisfying $\text{rank}(\Sigma) = \text{rank}(A)$ such that $A = U\Sigma V^*$ is the singular value decomposition of A.

(6 marks)

Question 4

Let V denote the vector space of polynomials of degree at most 2 over $\mathbb{R}$. Let

$$\mathcal{B} = \{1 + x, x + x^2, 1 + x + x^2\}$$

be a basis of V, and let $\mathcal{B}^* = \{f_1, f_2, f_3\}$ be the dual basis of $\mathcal{B}$.

(a) Find a matrix $D = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{pmatrix}$ such that

$$f_1(a + bx + cx^2) = d_{11}a + d_{12}b + d_{13}c$$

$$f_2(a + bx + cx^2) = d_{21}a + d_{22}b + d_{23}c$$

$$f_3(a + bx + cx^2) = d_{31}a + d_{32}b + d_{33}c$$

for every polynomial $a + bx + cx^2 \in V$.

(5 marks)

(b) Let W be the subspace of V spanned by the polynomial $1 + 2x + 3x^2$, and let W^0 be the subspace of V^* that annihilates W. If $\mathcal{C} = \{g_1, g_2\}$ is a basis of W^0, express g_1 and g_2 as linear combinations of f_1, f_2, f_3.

(8 marks)

(c) Let h_1, h_2, h_3 be linear functionals in V^* satisfying

$$h_1(p(x)) = p(-1),\ h_2(p(x)) = p(0),\ h_3(p(x)) = p(1)$$

for any polynomial $p(x) \in V$. Prove that $\mathcal{D}^* = \{h_1, h_2, h_3\}$ is a basis of V^* and find a basis $\mathcal{D}$ of V such that $\mathcal{D}^*$ is the dual basis of $\mathcal{D}$.

(7 marks)

Question 5

Let V be a complex inner product space and let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be an orthonormal basis of V.

(a) Prove that for any $x \in V$,

$$x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i.$$

(5 marks)

(b) For any $x, y \in V$, prove that

$$\langle x, y \rangle = \sum_{i=1}^{n} \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

(5 marks)

Question 6

Let

$$A = \begin{pmatrix} -1 & 3 & 2 \\ 3 & 1 & 4 \\ 2 & 4 & 1 \end{pmatrix}$$

be the matrix representation of a symmetric bilinear form f on $\mathbb{R}^3$ relative to the standard basis. Find an invertible matrix Q such that QAQ^T is a diagonal matrix, and determine the rank and signature of f.

(10 marks)

Question 7

Let A be a complex $n \times n$ matrix, and let A^* denote the conjugate transpose of A.

(a) Show that the nullspace of A^*A is equal to the nullspace of A.

(6 marks)

(b) If $A^*A = AA^*$, prove that the nullspace of A^* is equal to the nullspace of A.

(3 marks)

(c) Suppose that A has polar decomposition $A = UP$, where U is unitary, P is self-adjoint and non-negative. Prove that $A^*A = AA^*$ if and only if $UP^2 = P^2U$.

(6 marks)

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